

Role of cyclooxygenase isoforms in gastric mucosal defence.



A complex system of interacting mediators exists in the gastric mucosa to strengthen its resistance against injury. In this system prostaglandins play an important role. Prostaglandin biosynthesis is catalysed by the enzyme cyclooxygenase (COX), which exists in two isoforms, COX-1 and COX-2. Initially the concept was developed that COX-1 functions as housekeeping enzyme, whereas COX-2 yields prostaglandins involved in pathophysiological reactions such as inflammation. In the gastrointestinal tract, the maintenance of mucosal integrity was attributed exclusively to COX-1 without a contribution of COX-2 and ulcerogenic effects of non-steroidal anti-inflammatory drugs (NSAIDs) were believed to be the consequence of inhibition of COX-1. Recent findings, however, indicate that both COX-1 and COX-2 either alone or in concert contribute to gastric mucosal defence. Thus, in normal rat gastric mucosa specific inhibition of COX-1 does not elicit mucosal lesions despite near-maximal suppression of gastric prostaglandin formation. When a selective COX-2 inhibitor which is not ulcerogenic when given alone is added to the COX-1 inhibitor, severe gastric damage develops. In contrast to normal gastric mucosa which requires simultaneous inhibition of COX-1 and COX-2 for breakdown of mucosal resistance, in the acid-challenged rat stomach inhibition of COX-1 alone results in dose-dependent injury which is further increased by additional inhibition of COX-2 enzyme activity or prevention of acid-induced up-regulation of COX-2 expression by dexamethasone. COX-2 inhibitors do not damage the normal or acid-challenged gastric mucosa when given alone. However, when nitric oxide formation is suppressed or afferent nerves are defunctionalized, specific inhibition of COX-2 induces severe gastric damage. Ischemia-reperfusion of the gastric artery is associated with up-regulation of COX-2 but not COX-1 mRNA. COX-2 inhibitors or dexamethasone augment ischemia-reperfusion-induced gastric damage up to four-fold, an effect abolished by concurrent administration of 16,16-dimethyl-PGE(2). Selective inhibition of COX-1 is less effective. Furthermore, COX-2 inhibitors antagonize the protective effect of a mild irritant or intragastric peptone perfusion in the rat stomach, whereas the protection induced by chronic administration of endotoxin is mediated by COX-1. Finally, an important function of COX-2 is the acceleration of ulcer healing. COX-2 is up-regulated in chronic gastric ulcers and inhibitors of COX-2 impair the healing of ulcers to the same extent as non-selective NSAIDs. Taken together, these observations show that both COX isoenzymes are essential factors in mucosal defence with specific contributions in various physiological and pathophysiological situations.